ME182 - Citrus Juicer Project

Project Report

Manual Lever Citrus Juicer

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Group 11B

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1. General Information

The one being considered is the citrus juicer that is operated by a lever (usually hand-operated), which is often used to get the juice out of the oranges, lemons, and limes among other fruits. It is produced so as to be convenient, mess-free, and to juice without electricity.

General Overview

This juicer has a vertical pressing system. A user puts half of a citrus fruit on the press that has a shape of a cone, and pulls down the long handle. This lever is a very hard action, squeezing the juice, but leaving the seeds and pulp behind.

Advantages

- No electricity required
- Makes fresh natural juice on the spot.
- Clean and maintain easily.
- Strong and long-lasting

Common Uses

- Making fresh orange juice for breakfast
- To draw lemon juice to cook or drinks.
- Making new drinks in the restaurant or at home.

It is a dependable, environmentally friendly kitchen appliance and is worth buying despite price increases whenever there is a season to prepare citrus recipes.

2. Product Photographs

Figure 1: Front view of the Manual Lever Citrus Juicer

Figure 2: Top view

Figure 3: Side view

Figure 4: Close-up angled view

Figure 5: Front view showing cone and base

Patent Research and History Development: Manual Lever Citrus Juicer.

1. Introduction

A manual juice squeezer is a mechanical product that is intended to squeeze juice by compressive power. It has a whole-time history, which incorporates the fundamentals of mechanics exceptional efficiency, durability, and ergonomic advancements through patents.So firstly we will start by the beginning of the idea of squeezing the juice: It was an idea started by the Neothilic age (after the stone age 1500BC), and those people was trying to get some kinds of new stuff so they made a stone-bowl and a thick-stone smasher. They started to smash the fruits and the vegetables and mixing them trying to get a new flavor. Unfortunately, there are no special mechanism to do so they were so casual and it was hard to get the true flavor.

2. Juice Squeezer Early History (Pre-1900s -Early 1900s) The Inventions And Innovations Started From There

Prior to formal patents, juice was extracted by the use of:

- Wooden or simple wooden presses.
- Simple screw presses inspired by wine and olive Presses.

Mechanization:

Late 19th century: Cast metal squeegees were introduced.

- At the beginning of the 20th century: Reorientation towards the mechanized systems of pressing.

Key Patent Milestone:

- US1180959A – Fruit Press (1916)
  - Mechanical pressing introduced: plungers and linkages.
  - The start of engineered systems of juice extraction.

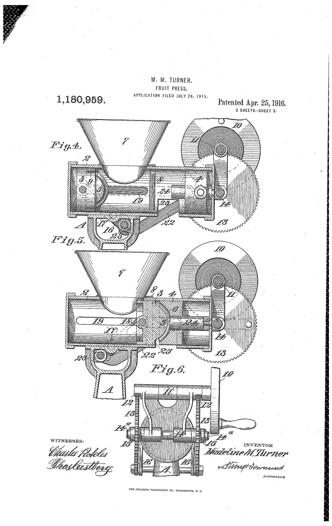

Figure 6: Patent drawing of US1180959A – Fruit Press (1916), M. M. Turner

The invention of Lever-Based Juicers (1920s–1950s) became developed.

During this time there was a lot of invention of designs that were closely related to your product.

Example Innovation:

Zaksenberg squeezer Juicer 1920s.

- Manual press, first heavy-duty manual lever.
- Made from cast iron for high strength

Engineering Advancements:

- Introduction of mechanism of lever to get mechanical advantage.
- Much higher endurance and reliability.

Mechanical Principle Introduced:

Lever systems enabled users to create compressive force with a small amount of effort.

4. Mid-Century squeeze juicer Improvements (1950s–1980s)

Patents during this time were geared towards efficiency and usability.

Key Patent:

- US2588906A – Citrus Fruit Press (1952)

Other Developments:

- Addition of filters/strainers
- Improved cone geometry
- Enhanced stability of base

Engineering Focus:

- Ideal force transmission.
- Reduction of energy loss
- Improved user ergonomics

Figure 7: Mid-century citrus press representative of US2588906A – Citrus Fruit Press (1952)

5. Modern Upgrades (2000s–Present)

Recent patents used to upgrade/develop and not to invent the squeezer.

Examples:

- US20120260810A1 – Citrus Juicer
  - Ergonomic handles
  - Security locking systems.
- WO2016093709A1 – Juice Extractor
  - Combines cutting + pressing mechanisms
- Design Patent D951719
  - Concentrate on looks and product design.

Figure 8: Modern commercial lever citrus juicer representative of US20120260810A1 and WO2016093709A1

Pull Lever Citrus Juicer.

1. Problem Statement

Prior to the invention of mechanical citrus squeezers users had many issues:

- Poor efficiency in the extraction of juice by squeezing.
- Excessive physical activity to squeeze juice.
- Dirt and spill in hand squeeze.
- Fruit seed present in the juice, pulp present.
- Inability to find a dependable and consistent system of extraction of juice.

2. User Needs

To solve these issues, the manual lever citrus juicer would meet the following requirements:

1. Efficient Juice Extraction

2. Reduced Physical Effort

- The product should enable users to use a minimum of input force.
- By using a lever system (mechanical advantage)

3. Occupation Clean and Controlled.

- Avoid spilling or splashing of juice.
- Provide a direct flow system into a container

4. JU separation with Solid Residue.

- Remove:
    - Seeds
    - Excess pulp
- Make sure of clear, potable juice.

5. Stability and Safety

- The device must:
    - Stabilize when in operation.
    - Stop falling or slipping.

6. Strength and Life Cycle.

1) users need a product that:

- Resists repetitive loads.

7. easy to Maintain.

- Easy to operate and minimum training.
- Clean and assemble/Disassemble dissimilarly.

3. Target Users

- Homes
- Cafes/restaurants
- Juice vendors

Functional Definitions: Citrus Juicer with Manual Lever.

1. Overall Function

The main use of the manual citrus juicer is to:

Transform the force of human input to a concentrated force of compressing to squeeze juice out of citrus fruits productively and seclude undesired solids.

2. Functional Decomposition

The product work is based on a series of interdependent functions:

1. Input Force Application

The lever handle is pulled by the user.

This is the initiating energy into the system.

Type: Human mechanical input

2. Force Transmission

Force is transferred through the lever:

Pivot joints

Guided shaft or linkages.

Function: Pass motion and force between handle to pressing head.

3. Force Amplification

Lever system augments the force applied through mechanical advantage.

A large compressive force is generated out of a small input force.

Engineering Role: Make it more efficient and less user effort.

4. Guided Linear Motion

The vertical shaft guarantees:

Straight downward movement

Correct positioning of parts.

Purpose: Be steady and accurate.

5. Fruit Compression

The pushing head compels the fruit on the cone.

There is compressive stress on fruit and juice is released.

Basic physical procedure: Mechanical deformation.

6. Juice Extraction

The leakage of the juice is caused by:

Applied pressure

Natural fluid movement

Function: Turn internal liquid into output.

7. Filtration / Separation

Built-in strainer separates:

Juice → passes through

Seeds & pulp → retained

Purpose: Enhance quality of output.

8. Juice Collection & Output

The directions of juice are:

Funnel

Spout

Use: To pour juice into a container, clean, and neat.

9. Structural Support

Base and frame:

Absorb reaction forces

Maintain stability

Control: Provide safe and reliable performance.

3. Functional Flow Summary

Input → Process → Output

Input: Human force

Action: Transmission - amplication - compression - filtration.

End product: Fresh, filtered citrus juice

4. Representation of functional block (text form).

User Force

↓

Lever Mechanism

↓

Force Amplification

↓

Guided Pressing Motion

↓

Fruit Compression

↓

Juice Extraction

↓

Filtration

↓

Juice Output

5. Supporting Functions

In addition to the main function, the product also performs:

Ergonomics holding of the object: comfortable grip and operation.

Safety control → avoids falling or any abrupt movement.

Cleanability, or maintainability → components that can be clean or remove.

6. Engineering Interpretation

As a mechanical engineering system, the juicer is a system that:

Works with simple machines (lever)

Applies compressive stress

Regulates the fluid flow and separation.

Holds body together during a stress.

7. Conclusion

The manual citrus juicer is a system of force transformation and fluid extraction that effectively transforms human inputs to useful output using a series of pre-determined mechanical functions.

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